

Sanket Lokegaonkar — Senior Applied Scientist

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Summary

Senior Applied Scientist focused on foundation model training. Co-led data strategy and training methodology for Amazon Rufus—a large-scale conversational AI shopping assistant serving hundreds of millions of customers—from data curation and scaling laws to training recipes, data mixture/curriculum, and productionization, shipping 5+ model generations with SOTA results on key benchmarks. Publications at ICLR, NAACL, SIGGRAPH, and IEEE BigData.

Research Interests

Large Language Models, Foundation Model Training, Data-Centric Machine Learning, Agentic Mid-Training, Scaling Laws, Representation Learning, Deep Learning

Experience

Amazon Rufus

Seattle, WA

Senior Applied Scientist

2021–Present

Founding member of the pre-training data team for Amazon Rufus, a conversational AI shopping assistant powered by frontier LLMs and serving hundreds of millions of customers. Co-led data strategy, and data mixture/curriculum design across 5+ model generations (dense and MoE). Previously built DataKit—distributed dataset management and checkpointing infrastructure for large-scale model training—as MLE-II (2021–2023).

- *Data curation*: Designed and scaled domain-specific data selection pipelines across shopping, STEM, math, and code—each with tailored methodology: hybrid retrieval-classifier for broad domains, two-stage extraction-classifier preserving LaTeX notation for math, and ensemble classifiers for code. Iterated across successive model generations with order-of-magnitude scale improvements and SOTA results on key benchmarks.
- *Training recipes*: Led batch size scaling and LR schedule ablation experiments that challenged conventional scaling law methodology—showing benchmark-optimal hyperparameters diverge from validation-loss-optimal. Designed multi-stage training curricula across pre-training, mid-training, and annealing phases. Findings directly shaped production training configuration for future model generations.
- *Agentic mid-training*: Leading data curation for agentic mid-training—instilling planning, tool-use, and autonomous decision-making capabilities into foundation models through continual pre-training.
- *Representation learning*: Led behavioral embeddings research—training LLMs to consume learned embeddings as a native input modality with curriculum-based reconstruction training for dense and MoE architectures. Developed web-graph significance ranking for long-context document stitching.
- *Data taxonomy*: Designed multi-dimensional data taxonomy (cognitive complexity, reasoning depth, educational level, timeliness) for compositional filtering of web-scale corpora. Distilled LLM annotations into lightweight classifiers for efficient inference at scale. Taxonomy-filtered subsets of lower-tier web data consistently outperformed the highest-quality curated baselines—significantly expanding the usable training corpus and shaping production data mixtures across training stages.
- Mentored 2 PhD interns → ICLR 2026 and NAACL 2025 publications.

Amazon Web Services

Seattle, WA

Software Development Engineer

2020–2021

Scaled distributed configuration tracking and change management services for AWS Config, supporting compliance auditing and resource inventory across millions of cloud resources.

Uber Advanced Technologies Group

Pittsburgh, PA

Software Engineer

2019–2020

Integrated learned behavior prediction models into the motion planning stack for autonomous vehicles. Developed autoregressive probabilistic flow models for multi-agent trajectory forecasting in complex urban environments.

Bloomberg L.P.

New York, NY

Software Engineer

2018–2019

Built real-time distributed rules engine in Spark/Storm for regulatory transaction monitoring, processing thousands of transactions per minute. Developed algorithmic trading analytics workflows for Bloomberg Terminal.

Virginia Tech – Vision & Learning Lab

Research Assistant

Advisor: Prof. Jia-Bin Huang

- MS thesis on continual learning for semantic segmentation: mitigating catastrophic forgetting via weight regularization and knowledge distillation.
- Built photogrammetry pipeline for crowdsourced 3D reconstruction (SIGGRAPH'19).

Blacksburg, VA

2017–2018

Discovery Analytics Center, Virginia Tech

Research Assistant

Advisor: Prof. Naren Ramakrishnan

- Developed driver state detection combining face tracking, vehicle telemetry, and recurrent models on naturalistic driving data (IEEE BigData'18).

Blacksburg, VA

Summer 2017

Education

Virginia Tech

M.S. Computer Science

Thesis: Continual Learning for Deep Dense Prediction. Advisor: Prof. Jia-Bin Huang

Blacksburg, VA

2016–2018

University of Mumbai

B.E. Computer Engineering

Mumbai, India

2011–2015

Publications

ICLR '26: C. Deng, S. Lokegaonkar, C. Lockard, B. Fetahu, N. Zalmout, X. Li.

ByteFlow: Language Modeling through Adaptive Byte Compression without a Tokenizer

NAACL '25: Y. Zhuang, J. Yang, H. Jiang, X. Liu, K. Cheng, S. Lokegaonkar, et al.

Hephaestus: Improving Fundamental Agent Capabilities of LLMs through Continual Pre-Training

2025: J.Y. Lee, H. Bonab, N. Zalmout, M. Zeng, S. Lokegaonkar, et al.

DocTalk: Scalable Graph-Based Dialogue Synthesis for LLM Conversational Capabilities

2025: W. Wang, R. Lin, S. Li, C. Lockard, R. Sarkhel, S. Lokegaonkar, et al.

UniFilter: Unified Multimodal Data Quality Classifier with Synthetic Data

SIGGRAPH '19: S. Blanchard, J.B. Huang, C.B. Williams, V. Meenakshisundaram, J. Kubalak, S. Lokegaonkar.

SourceForm: An Automated Crowdsourced Object Generator

BigData '18: S. Yadawadkar, B. Mayer, S. Lokegaonkar, M.R. Islam, N. Ramakrishnan, et al.

Identifying Distracted and Drowsy Drivers Using Naturalistic Driving Data

IJCAI '16: S.S. Shah, S. Bhattad, S. Lokegaonkar, G. Ramakrishnan.

Building Complementary Domain Taxonomies using Query Enrichment

Workshop

Technical Skills

ML: PyTorch, large-scale distributed LLM training, MoE architectures, vLLM

Data: Spark, web-scale data curation and processing pipelines

Languages: Python, C/C++, Java, SQL